1. Student Details

- **Name:** Apurav Kashyap
- **Roll Number:** 24f2000063
- **Email:** 24f2000063@ds.study.iitm.ac.in
- **About Me:** I am a student at the IIT Madras BS Degree program with a strong focus on the logical "thinking mechanisms" behind web development and algorithms. I enjoy building structured, user-centric applications and have a keen interest in combining philosophy, human behavior, and tech.

2. Project Details

- **Project Title:** Placement Management System
- **Problem Statement:** To design and build a secure, role-based web application that streamlines the campus recruitment process, allowing companies to post jobs, students to apply seamlessly, and administrators to oversee the entire pipeline without data clutter.
- **Approach:** The application was built using the Flask web framework following the MVT (Model-View-Template) architecture. Strict Role-Based Access Control (RBAC) was implemented to ensure security and logical data isolation between Admins, Companies, and Students. Focus was placed on intuitive UI/UX using Glassmorphism and intelligent data filtering (like auto-expiring job drives) to provide a real-world SaaS experience.

3. AI/LLM Declaration

I used AI tools to assist in debugging specific logic flows, formatting the Bootstrap 5 UI/UX (Glassmorphism CSS), and understanding advanced SQLAlchemy relationship syntax. The extent of AI/LLM usage is around 20-30%, limited to styling, syntax correction, and documentation formatting. The core logic, routing, thinking mechanisms, and final integration were designed and implemented manually

4. Technologies and Frameworks Used

Technology / Library	**Purpose**
Flask	Core backend web framework
SQLAlchemy	Object Relational Mapper for SQLite database
Jinja2	Template engine for rendering dynamic HTML pages
Bootstrap 5 & Custom CSS	Frontend styling, responsive design, and Glassmorphism effects
Flask-Login	User authentication, session management, and RBAC
WTForms	Frontend and backend form validation
Werkzeug.security	Secure password hashing
SQLite	Lightweight local database for storing application data

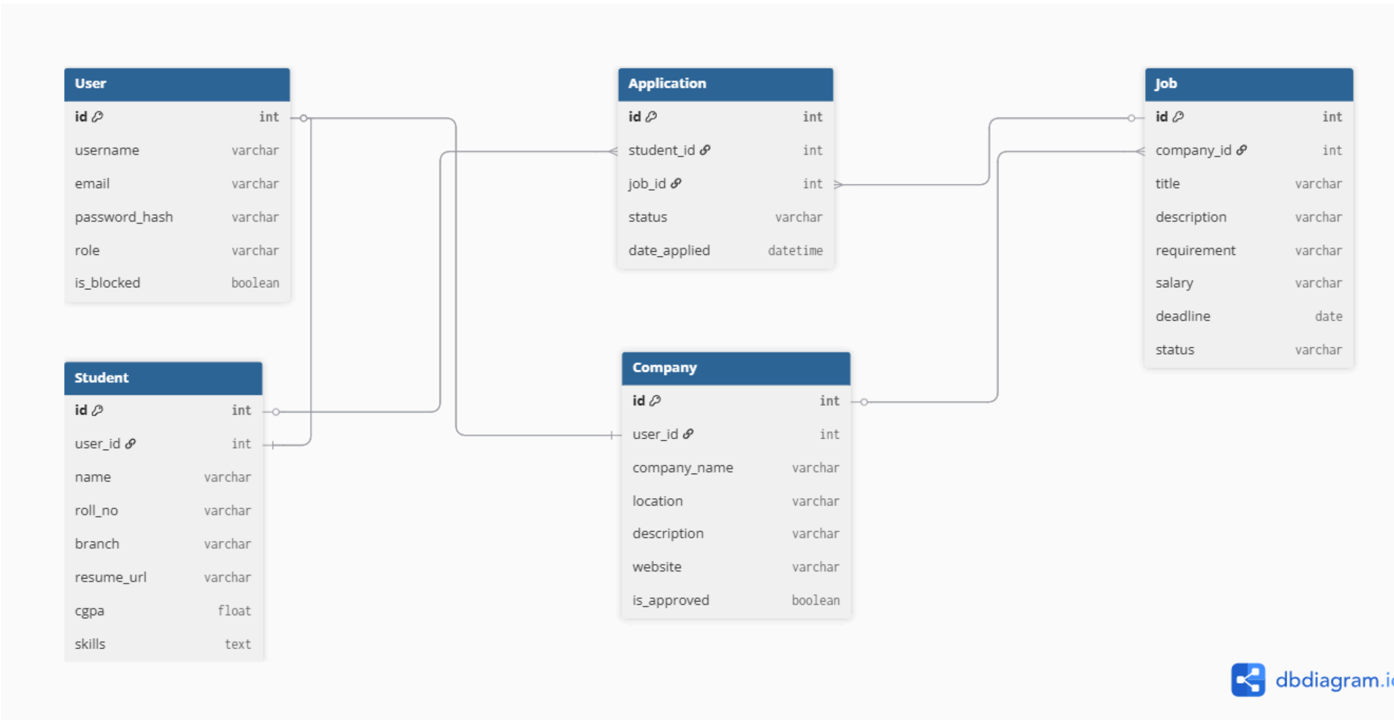
5. Database Schema / ER Diagram

Tables:

1. **User:** Stores authentication credentials and role flags (id, email, password, role).
2. **StudentProfile:** Stores student-specific details (id, user_id, name, cgpa, skills).
3. **CompanyProfile:** Stores company details (id, user_id, company_name, industry, is_approved).
4. **Job:** Logs placement drives created by companies (id, company_id, title, salary, deadline, is_active).
5. **Application:** A junction table logging student applications for jobs (id, student_id, job_id, status, date_applied).

Relationships:

- **One-to-One:** User → StudentProfile
- **One-to-One:** User → CompanyProfile
- **One-to-Many:** CompanyProfile → Job
- **Many-to-Many (via Application table):** StudentProfile ↔ Job



7. Video Presentation

- **DriveLink:**https://drive.google.com/file/d/1QQohvdWmJoPCtgA8p68ZNjegf34qyLLG/view